

BHARATH KUMAR KAGEY

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Senior Mechanical Engineer | HPDC Tooling & Manufacturing Systems

SUMMARY

9+ years of OEM mechanical design experience with 6 years in HPDC tooling at Tesla, leading die development programs from concept through high-volume production. Specializes in Castings Gating design thermal system design, die longevity architecture, simulation-driven development, and PLM implementation across GIGA-scale casting programs. Experienced in cross-factory collaboration, supplier technical control, and engineering tool automation.

ENGINEERING EXPERIENCE

Tesla Motors — Fremont, CA

July 2020 – Current

Sr. Tooling Engineer – High Pressure Die Casting | Acting Engineering Supervisor

- Designed runner, gating, and venting systems for HPDC dies including Torque Box and Rear Node dual-cavity programs using NADCA and Buhler equations, reducing Torque Box full shot weight by 3.5 kg and cycle time from 85 sec to 68 sec.
- Performed MAGMASOFT flow and thermal simulations for Lathrop Model S/X dies, optimizing runner geometry, cooling circuit layout, and thermal balance to reduce cycle time, die soldering, and casting defects.
- Introduced die longevity technologies to GIGA Castings tooling fleet — 3D-printed conformal cooling inserts, advanced steel alloys, controlled surface coating, and heat treatment specifications — extending die service life and reducing unplanned maintenance intervals.
- Acted as Engineering Supervisor for GIGA Castings (Jan 2023 – Jul 2025), managing 5 Tool & Die technicians across two 6200-ton DCMs and leading production launches for RUB, Torque Box, Shock Tower, and Rear Node GIGA dies.
- Designed machined 6061 aluminum manifold bodies with brass quick-connect couplers and O-ring seals to replace stainless steel hard plumbing on GIGA dies — reducing die refurbishment turnaround from 6 weeks to 2 weeks and subsequently adopted as the standard cooling interface on the FRUB die program.

Tools: CATIA V5, MAGMASOFT, ANSYS Static Structural, 3DEXPERIENCE / ENOVIA, PowerShell, AI-assisted scripting

Segula Technologies (Volvo Trucks NA) — Greensboro, NC

July 2018 – May 2020

Design Engineer – CAB | Geometrical Assurance Engineer | Fixture Design Engineer

- Maintained CAB design for Volvo and Mack Trucks fleet across early and late-stage releases, resolving factory design issues and managing engineering change notices for structural and design releases in CATIA V5 and Windchill PLM.
- Led front steel bumper design for Mack Anthem from concept through production release — designed mounting brackets, front plastic deflectors, and all installation and assembly requirement drawings.
- Performed Geometrical Assurance Engineering (GAE) on New VNL chassis fairing brackets and wiper assembly using RDnT — generated datum schemes, ran RSS tolerance stack-up calculations, and issued 3D/2D design recommendations for hole sizes and slot locations.
- Prepared pre-tolerance documents ahead of DCN freeze, created measurement point documents for prototype and production-level part inspection, and collaborated with fixture designers to control assembly sequence for dimensional stability.
- Designed production fixtures for Mack Macungie Plant lines including engine rack, turbo pipe, and floor mat drilling fixtures; reverse engineered existing fixtures for high-volume manufacturing.
- Managed new part creation, variant documentation, and structural releases in KOLA across hood plenum, HVAC panel, upper bunk brackets, and BIW components on multiple truck platforms.

Tools: CATIA V5, PTC Windchill PDM Link, RDnT, Creo View Lite, UG-NX

Primark Infotech — India

Feb 2014 – Dec 2015

Product Designer

- Designed plastic injection molded components for automotive CAB interior trim in CATIA V5, covering draft analysis, parting line design, clash analysis, and DFM/DFA validation from concept through production release.
- Generated detailed GD&T drawings, managed BOM, performed DRBFM failure mode analysis, and supported material selection across multiple interior trim subsystems.

PROJECTS LED

Runner Design & Thermal Optimization — Torque Box and Rear Node (Model S/X)

- Designed the complete runner, gating, and venting system for the Rear Node dual-cavity die from scratch using NADCA and Buhler equations and led the project through trim die buy-off into production.
- Redesigned the Torque Box runner and gating system to resolve chronic air entrapment defects, reducing full shot weight by 3.5 kg and cycle time from 85 sec to 68 sec — validated across multiple design concepts in MAGMASOFT before committing to steel changes.
- Ran MAGMASOFT flow simulations across candidate runner designs to evaluate melt velocity, temperature distribution, pressure, and fill trace at each gate — selecting the optimal design based on simulation before any physical trials.
- Simulated die thermal system for both programs in MAGMASOFT to optimize cooling line layout, reduce die soldering, and improve thermal balance across cover and ejector halves.
- Performed ANSYS structural analysis on cover and ejector halves and DCM platens to correctly size support pillar preload, reducing die deflection under shot pressure and eliminating flash at parting surfaces.

FRUB Die Development — 200,000 Shot Die Life Program

- Led FRUB HPDC die development from scratch targeting 200,000 shot die life, up from 140,000 shots on the previous die, and reduced cycle time from 105 sec to 85 sec through coordinated improvements across die design, cooling, materials, and gating.
- Redesigned die architecture with a dedicated insert holder and sub-inserted cavity blocks, replacing monolithic inserts with modular replaceable sections — combined with quick-connect 6061 aluminum cooling manifolds and couplers to eliminate traditional hard plumbing and cut die refurbishment turnaround from weeks to days.
- Reworked runner and gating to fix chronic fill issues in the motor cover and wheel house areas causing casting cracks and scraps ran MAGMASOFT flow and solidification simulations and ANSYS stress analysis to validate the design before cutting steel.
- Traveled to Giga Berlin to review their FRUB die experience firsthand, incorporated their lessons learned and prior improvement trials into the new design to avoid repeating known failure modes.
- Worked directly with steel suppliers to define surface coating treatment and heat treatment parameters zone-by-zone across insert sections, improving surface hardness at high-wear gating and overflow areas — combined with BTU-calculated conformal cooling layout to reduce thermal cycling stress and extend insert service life toward the 200k shot target.

3DEXPERIENCE PLM Implementation — GIGA Castings Tooling (Lathrop & Fremont)

- Led migration of GIGA Castings tooling CAD and documentation from OneDrive to 3DEXPERIENCE PLM across Lathrop and Fremont sites — establishing structured, revision-controlled data management for the first time on the tooling team.
- Authored CAD standards and published formal technical documentation for GIGA tooling design practices, giving the team a consistent reference for CAD structure, naming conventions, and release procedures.
- Designed and configured a standard component library within 3DX to standardize common die components and tolerances across programs, reducing redundant design work and improving reuse across die families.
- Created a dedicated collaborative space for Castings Tooling within 3DX and customized the design release workflow to fit tooling-specific requirements — replacing ad hoc file sharing with a structured, auditable release process.
- Worked directly with Dassault to resolve data migration challenges and manage large tooling dataset transfer into 3DX, ensuring data integrity and accessibility for collaborative design across both sites.

GD&T Standards Development — Tesla GIGA Die Families

- Authored the GD&T and datum scheme standard for Tesla GIGA die tooling, establishing tolerance requirements accounting for three competing demands simultaneously: in-production thermal expansion, flash prevention at parting surfaces, and insert interchangeability during refurbishment.
- Applied RSS statistical tolerance analysis to determine probable assembly variation, avoiding unnecessarily tight manufacturing tolerances that inflate tooling cost without quality benefit.
- Defined insert fit tolerances and locating datum schemes enabling drop-in replacement of worn inserts during die refurbishment, saving 2 hours per insert replacement in the tool shop.
- Integrated standard into ENOVIA/PDM as the governing reference for all GIGA die releases, ensuring consistent supplier drawing interpretation and reducing revision cycles during die buy-off.

Automated Sub-Insert Drawing Generation — PowerShell / Claude CODE Pipeline

- Independently developed a 3D-to-2D drawing automation pipeline using PowerShell and CATIA with AI-assisted code generation, converting sub-insert CAD models directly into dimensioned, GD&T-annotated production drawings.
- Defined all engineering requirements driving the automation — datum schemes, GD&T callouts, tolerance values, title blocks, and revision tables — based on the internally developed GIGA die GD&T standard.
- Automated generation of sub-insert drawings and ignition schematics covering 20–50 drawings per die set, reducing per-drawing turnaround from up to 2 hours to near-instant output. Deployed to supplier drawing releases as production-quality deliverables.

Engineering Supervisor — GIGA Castings, Model Y (Jan 2023 – Jul 2025)

- Acted as Engineering Supervisor for GIGA Castings from Jan 2023 through Jul 2025, managing a team of 5 Tool & Die technicians across two 6200-ton DCMs on 12-hour Pittman shifts as the on-floor SME for all tooling-related production issues.
- Led die validations and production launches for RUB, Torque Box, Shock Tower, and Rear Node GIGA dies into high-volume Model Y production, driving process parameter optimization across shot profile, sprayer settings, and die temperature management to improve casting yield.
- Implemented advanced die longevity technologies including 3D-printed conformal cooling inserts, Die-Therm cooling systems, and novel steel alloy trials — collaborating with cross-functional teams across Die Design, Manufacturing, Quality, and Safety to resolve systemic issues affecting line uptime and part quality.

EDUCATION

Master of Science, Automotive Engineering — Lawrence Technological University

December 2017

Bachelor of Science, Automotive Engineering — Osmania University

May 2015